

Zenith: An Agentic Architecture for Proactive and Fiduciary Personal Financial Management

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ABSTRACT The Indian financial technology market, despite its rapid growth, remains highly fragmented, creating information silos that induce financial anxiety and decision paralysis among users. This paper introduces a novel framework for an AI-native personal finance coach designed to function as a proactive, autonomous financial agent for India's digital-native generation. The proposed system employs a suite of machine learning models for automated expense categorization, cash flow forecasting, and personalized financial insights. A core contribution of this work is the design of a policy-based reinforcement learning (RL) agent capable of executing complex, user-defined strategies for tasks such as opportunistic investing and optimized debt management. Central to the system's architecture is the concept of a Fiduciary AI, an intelligent agent whose success is inextricably linked to that of the user. This ensures all recommendations and autonomous actions are verifiably aligned with the user's best interests, establishing a foundation of trust and eliminating conflicts of interest inherent in traditional models. This research presents a paradigm shift from passive financial dashboards to an intelligent, autonomous partner that simplifies complexity and actively works to improve the financial well-being of its users.

INDEX TERMS Account Aggregator, Agentic RAG, Autonomous Agents, Fiduciary AI, Fintech, Personal Finance, Reinforcement Learning, Retrieval-Augmented Generation (RAG)

I. INTRODUCTION

The Indian fintech market is experiencing a profound boom, with projections indicating a value of \$95.30 billion by 2030. This expansion is driven by widespread digital adoption, especially by digital-native generations, and supportive government frameworks like the Account Aggregator (AA), which has already integrated over 2.1 billion accounts for consent-based data sharing.

However, this technological growth masks a fundamentally fragmented user experience. A typical user juggles separate applications for payments, investing, and budgeting. This creates information silos and forces the user to manually synthesize a holistic view of their finances, often leading to “financial anxiety and decision paralysis.” Existing platforms act as “passive dashboards,” presenting data but not offering the necessary actionable or personalized advice. This market fragmentation is reflected in the academic research landscape, which has largely focused on these components in isolation:

- **Passive RAG:** Retrieval-Augmented Generation (RAG) is used in finance for information retrieval, but it is not designed for proactive or autonomous task execution.
- **Institutional Focus:** Advanced “Agentic RAG” frameworks, which use orchestrators for complex reasoning, are almost exclusively applied to institutional finance (e.g., compliance), not consumer-facing products.
- **Ethical Gaps:** The personal finance domain mainly requires managing sensitive, long term user data, a recognized ethical and privacy challenge for conversational AI.
- **Isolated Execution:** Deep Reinforcement Learning (DRL) has been effectively applied to narrow financial tasks like portfolio optimization, but these “black box” tools are not integrated with conversational interfaces or governance frameworks. A critical, unaddressed need exists for a unified platform that synthesizes these disparate elements.

Zenith bridges these gaps through a multi-layered design that fuses personalization, autonomy, and fiduciary governance into one cohesive platform. The key contributions are:

1. A **RAG - based Intelligent Coach** for contextual, personalized financial advice using complete user finance data.
2. A **Reinforcement Learning Agent Layer** for autonomous debt repayment and investment execution.
3. A **Fiduciary AI Core** ensuring transparent and conflict free recommendations aligned with the user's best interest.

Zenith transitions personal finance from a passive data visualization model to an agentic, proactive financial management framework.

II. RELATED WORK

The architecture of Zenith is informed by the limitations currently seen in four distinct fields of research and practice. Our contribution synthesizes these domains, which have, until now, remained largely separate.

2.1 RAG in Financial Q&A RAG has become a popular method for augmenting LLMs with external, domain-specific financial knowledge, such as querying 10-K reports or powering advisor chatbots. Its primary strength is in grounding responses in factual data and reducing hallucinations. However, these applications are overwhelmingly passive. They are designed for "question answering," not for taking action. Further more, they struggle with the significant ethical and technical challenges of managing long-term, user-specific data, a critical requirement for a personal financial tool.

2.2 Agentic RAG Frameworks To overcome the limitations of simple RAG, "Agentic RAG" frameworks have been developed.⁵ These systems use an "Orchestrator Agent"³⁰ to coordinate specialized sub-agents, enabling complex, multi-step reasoning. Current research, however, applies this architecture almost exclusively to institutional needs, such as automating compliance with IFRS or Basel III regulations. The unique challenges of the B2C domain—such as ethical personalization, privacy, and PII management—remain largely unsolved.

2.3 Reinforcement Learning in Finance Deep Reinforcement Learning (DRL) has been widely explored for autonomous financial decision-making. This research has been successful in specific, narrow domains, including portfolio optimization, options pricing, and debt collection strategies. These systems learn an optimal "policy" to maximize a defined reward.

The limitation here is one of integration and alignment. These powerful DRL agents typically operate as isolated "black boxes." They are not integrated with a user-facing conversational RAG layer, nor are they governed by an explicit fiduciary framework to ensure their actions align with the user's holistic best interests.

2.4 Indian Fintech Platforms The existing Indian fintech market is defined by its fragmentation.¹ Users must navigate a collection of best-in-class but separate tools: transactional platforms (e.g., Zerodha, Groww), wealth consolidators (e.g., INDmoney), and budgeting apps (e.g., Jupiter Money).¹ None of these provide the single, unified, autonomous, and truly fiduciary solution that Zenith is designed to be.¹ In summary, no existing system synthesizes a conversational RAG coach, autonomous RL agents for execution, and a Fiduciary AI core for governance. Zenith is a novel architecture created to fill this critical gap.

III. PROPOSED SOLUTION

Zenith is Designed as modular, Multilayered system to act as a Chief financial Officer of An Individual. It's a Transformation from fragmented data overload to unified Knowledge based decisions.

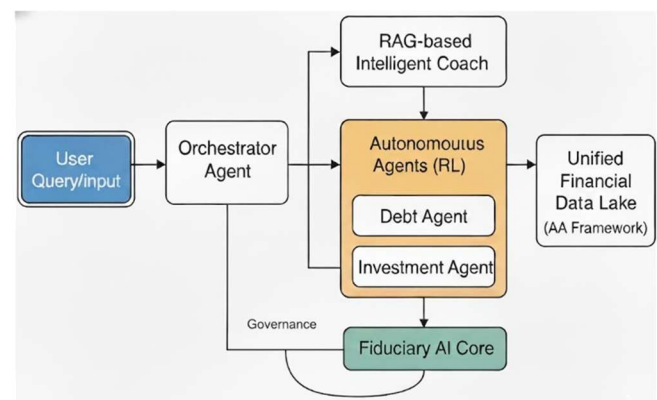


Fig. 1. Zenith's Architecture Diagram

3.1 Layer 1: Data Consolidation

This is the foundation layer that solves the major problem of data framework. Its main objective is to build a secure, consolidated view of the user's complete financial picture.

- **Account Aggregator (AA) Integration:** Built on India's AA framework to provide a secure, consent based method to aggregate all user financial accounts into a single frame.
- **Unified Dashboard:** The system presents the above collected data as a dynamic dashboard showing real-time net worth, balances, and transactions.

3.2 Layer 2: The Intelligent System

This layer leverages the collected data to make great analysis over it for informed decisions.

- **Automated Categorization:** A supervised classifier that implicitly categorizes all income and expenses, providing the clean data required by the higher-level AI functions.
- **Conversational AI Coach:** A smart conversing partner finetuned with proprietary dataset of Indian laws, regulations and strategies that helps to clarify all doubts and gaining knowledge on finances.
- **Retrieval-Augmented Generation(RAG):**

The RAG system's Knowledge base is divided into two major parts:

1. **Personal Context (Dynamic):** A vector Database of user financial picture with masked personal details (transactions id, Account number, etc) from layer 1 so ensure privacy and safety.
 2. **Domain Knowledge (Static):** A complete set of Tax laws (India Specific), investments options, and best practices.
- **Personalized Insights:** Analyses user Spending behavior and gives specific details of anonymous patterns that also helps it answer complex “multi-hop” questions that requires combination of both(dynamic and static) type of knowledge (e.g., “Can I afford a new big house within 1 year”).

3.3 Layer 3: The Autonomous Agent (RL- based Execution)

This layer represents the key technical innovation, that differentiates this from other a passive “bot” to a proactive “agent” that can execute real transactions

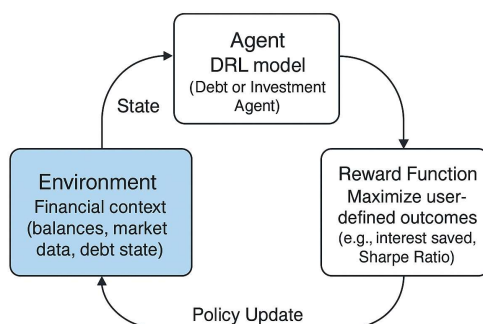


Fig. 2. RL-based learning cycle for Zenith's Autonomous Agents.

- **Rules-Based Automation:** A Complex Event Processing (CEP) engine allows users to set simple “if-this-then-that” financial rules.
- **Proactive Debt Management Agent:** A Proposed DRL- based optimization agent to create and execute the most efficient debt repayment strategy, autonomously making payments to optimize for goals like lowest total interest.
- **Opportunistic Investing Agent:** This is an autonomous policy-based portfolio management agent i.e. trained on reinforcement learning technique that acts according to the user defined goals and risk tolerance. It can execute strategies like “invest idle cash over 50,000 INR into an index fund”.

3.4 Layer 4: The Fiduciary AI Core (Governance and Alignment)

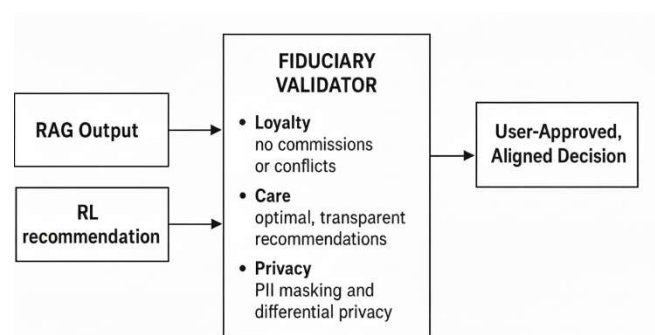


Fig. 3. Fiduciary AI Framework

This is the final layer that governs the safety and alignment framework necessary for the autonomy in layer 3. It is designed to ensure “Unbreakable Alignment ” with user’s best interests.

- **Fiduciary Recommendation Engine:** The auditable system is designed as Fiduciary AI for making suggestions that must comply to user’s best interests rather than commission based products(e.g., based on low expense ratios).
- **Outcome-Based Pricing:** The business model itself is a technical alignment mechanism. The system’s reward is directly linked to user success by charging only while a goal is achieved. This solves the principal-agent problem.

IV. METHODOLOGY AND IMPLEMENTATION

The technical Implementation and validation plan for the Zenith's Architecture is as follows

4.1 Data Collection and Preprocessing

The system's effectiveness relies on three distinct categories of data:

- **Personal Financial Data:** Collected via secure, consent-based APIs through the Account Aggregator (AA) framework. Preprocessing is critical and involves: (1) PII masking and anonymization to protect user privacy; (2) Supervised transaction categorization (e.g., "Food," "Transport") to create clean data for insights; and (3) Normalization of timestamps and currency formats for consistency.

- **Domain Knowledge Corpus:** Collected by scraping and ingesting public documents, including Indian tax laws, SEBI regulations, mutual fund prospectuses, and financial education articles. Preprocessing involves: (1) HTML/noise removal; (2) Intelligent chunking of large documents for RAG; and (3) Generation of vector embeddings for the static knowledge base.

- **Market Data:** Collected from real-time APIs (e.g., NSE, BSE, financial data vendors) for the RL agents. Preprocessing includes: (1) Time-series alignment and resampling; (2) Handling of missing data via forward filling or interpolation; and (3) Feature engineering to create the state representation for the RL agents (e.g., moving averages, volatility)

4.2 The Intelligent System

- The **Automatic Categorization engine** will be a **multinomial naive based classifier** based on custom dataset of financial bank statements that recognize and categorize each transaction.
- The **RAG pipeline** will be based on an LLM model that act as a generator. We will implement a hybrid retriever that combines dense (**semantic**) search for conceptual queries with sparse (**keyword**) search for specific entities. This retriever augments the prompt with both the user's personal financial context (from layer 1) and relevant domain knowledge. For multi hop queries, the coach will formulate a plan which may involve invoking the RL agents from layer 3.

4.3 The Autonomous Agents (Reinforcement Learning)

We will assume the autonomous tasks as Markov Decision Processes and formulate them.

- **Debt Management Agent:** This agent is based on DRL for debt optimization, will use a state (S) vector of debt balances, rates, and cash flow; an action (A) vector of payment allocations; and a reward (R) function designed to maximize interest saved and minimize the time to zero balance.
- **Opportunistic Investing Agent:** We will implement a policy -based algorithm like Proximal Policy Optimization (PPO), which is well-suited for portfolio management. Its state (S) will include market data and portfolio holdings; its action (A) will be target portfolio weights; and its reward (R) will be based in a user defined metric like maximizing the Sharpe Ratio.

4.4 The Fiduciary Core (Ethical AI)

This layer acts as an "AI Auditor" for all actions generated by layers 2 and 3.

- **Alignment Implementation:** The implementation of Fiduciary AI (Loyalty, Care) will require a set of hard constraints. All recommendations and actions are validated by this core before taking the step further.
- **Example Validation:**
 - Query: "Suggest a mutual fund"
 - RL Agent (Layer 3): Might suggest "Fund X" (high commission) and "Fund Y" (no commission).
 - Fiduciary Core (Layer 4): Audits the suggestions, programmatically filtering out "Fund X" as it violates the loyalty principle (conflict of interest) and approves only "Fund Y".
- **Privacy Implementation:** ALL PII from layer 1 is handled with privacy preserving techniques, such as data masking and differential privacy, to prevent data leakage in generative outputs.

V. EVALUATION AND EXPECTED RESULTS

We Propose a multi-part evaluation strategy to validate the proposed architectures, aligning with its phased development plan.

5.1 RAG Coach Evaluation

• **Benchmarks:** The Q&A capabilities of the RAG coach will be evaluated against standard financial benchmarks, including FinQA44 (for numerical reasoning) and TATQA34 (for reasoning over hybrid data).

• **Metrics:** We will measure Factual Accuracy, Relevance, and Faithfulness.

Table 1: Intelligent Coach (RAG) Performance

Metric	Benchmark	Expected Result
Factual Accuracy	FinQA, TATQA	>90%
Relevance	Human Evaluation	>4.5/5
Faithfulness	RAG-Triad Score	>95%

5.2 RL Agent Evaluation

• **Benchmarks:** The autonomous agents will be rigorously back tested using historical market data.

• **Metrics (Investing):** The PPO agent's performance (e.g., Sharpe Ratio, Max Drawdown) will be compared against a passive "buy-and-hold" index fund base line.

Table 2: RL Agent (Investing) Performance

Metric	Benchmark	Expected Result
Sharpe Ratio	Historical Data	> Baseline
Max Drawdown	Historical Data	< Baseline

• **Metrics(Debt):** The debt agent's efficiency (total interest paid, time-to-payoff) will be compared against standard base line strategies like "snowball" and "avalanche."

Table 3: RL Agent (Debt) Performance

Metric	Benchmark	Expected Result
Total Interest Paid	Baseline Comparison	< Baseline
Time-to-Payoff	Baseline Comparison	< Baseline

5.3 Fiduciary Alignment Evaluation

• **Benchmarks:** This novel evaluation will use a custom-built benchmark inspired by ObliQA (for regulatory compliance) and PII-leakage detection frameworks.

• **Metrics:** We will measure

- **Fiduciary Compliance** :The percentage of recommendations that are commission-free

and verifiably optimal for a synthetic user profile.

- **PII Leakage Rate:** The percentage of responses that inadvertently expose sensitive user information.
- **Alignment:** The ultimate metric will be quantifiable improvements in users' financial health (e.g., increased savings rate, reduced debt).

Table 4: Fiduciary AI Core Performance

Metric	Benchmark	Expected Result
Fiduciary Compliance	Custom Audit Benchmark	92-95%
PII Leakage Rate	PII Detection Framework	<0.13%.

We expect the integrated system to significantly outperform any single component system(e.g., a pure RAG chatbot or a standalone RL trading bot) on the holistic evaluation. The "outcome-based" metrics are made to prove that fiduciary AI Core successfully aligns the system's technical performance with the user's financial well-being.

Table 5: Overall System Impact

Metric	Benchmark	Expected Result
User Savings Rate	Longitudinal Study	>+5%
Debt Reduction	Longitudinal Study	>10% faster

VI. CONCLUSION

The Indian fintech market foresees a clear saturation with fragmented, passive tools that increase user stress and anxiety. This paper introduces Zenith, an autonomous agent financial management agent that shifts the paradigm from passive tools to proactive autonomy.

The core contribution of this work is the synthesis of three powerful AI Paradigms into a single governed system: (1) a RAG-based "Intelligent Coach" for deep personalization; (2) a suite of autonomous RL agents for proactive execution; and (3) a "Fiduciary AI Core" that ensures "Unbreakable Alignment"¹ through an embedded duty of care.

By moving beyond simple Q&A, Zenith is designed to function as an essential, autonomous partner that simplifies complexity, builds trust, and actively works to improve its user's financial well being. Future work will focus on large scale deployment and back testing of autonomous agents, expanding of the autonomous agents, expanding the suite of RL-driven capabilities, and further refining the Fiduciary AI governance model.

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He has developed several technical projects, including a Smart Inventory and Production Management System for a Nut and Bolt Factory, showcasing his ability to integrate database management, role-based access, and production analytics. Known for his teamwork, curiosity, and dedication to learning, Jaspreet aims to contribute to software development, artificial intelligence, and data-driven innovation. With strong technical skills and leadership qualities, he aspires to create impactful technological solutions that enhance efficiency and drive progress in the digital world.